

- 6 The curve C has equation $y = \frac{x^2 + 2x - 15}{x - 2}$.

(a) Find the equations of the asymptotes of C . [3]

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(b) Show that C has no stationary points. [3]

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- (c) Sketch C , stating the coordinates of the intersections with the axes.

[3]

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- (d) Sketch the curve with equation $y = \left| \frac{x^2 - 2x - 15}{x - 2} \right|$.

[2]

- (e) Find the set of values of x for which $\left| \frac{2x^2 + 4x - 30}{x - 2} \right| < 15$. [4]

This image shows a full page of white paper with horizontal dashed lines, typical of primary school handwriting practice paper. The lines are evenly spaced and run across the entire width of the page. There are no margins, text, or other markings present.